

PROFILE

Computational Biologist and Ph.D. Researcher skilled in Explainable AI (XAI), Machine Learning, and Large Language Model (LLM) integration for biomarker discovery and multi-omics research. Authored multiple studies on respiratory and metabolic disorders, emphasizing interpretable AI, and contributed to a design patent advancing translational biotechnology innovation.

CONTACT

- Phone: +91-9810769156
- Email: nakultanwar_2k22phdbt507@dtu.ac.in
- LinkedIn: www.linkedin.com/in/nakul-tanwar-a714a4184
- ORCID ID: <https://orcid.org/0009-0006-2520-1177>

PUBICATIONS:

Journal Articles:

- Tanwar, N., Hasija, Y.** Explicating miRNA-mediated regulation of inflammatory pathways in COPD, MS, and lung cancer using explainable artificial intelligence: insights from peripheral blood profiles, *Integrative Biology*, Volume 17, 2025, zyaf020, <https://doi.org/10.1093/intbio/zyaf020>
- Tanwar, N., Hasija, Y.** Explicate molecular landscape of combined pulmonary fibrosis and emphysema through explainable artificial intelligence: a comprehensive analysis of ILD and COPD interactions using RNA from whole lung homogenates. *Med Biol Eng Comput* 62, 2557–2570 (2024). <https://doi.org/10.1007/s11517-024-03099-8>
- Tanwar, N., & Hasija, Y.**** (2025). An eXplainable AI and Conversational Decision Support System for Lung Cancer Diagnosis and Monitoring. *Technology and Health Care*. (Manuscript Under Review <https://zenodo.org/records/15570514>)

Book Chapters:

- Tanwar, N., & Hasija, Y.** (2024). Overview of new trends on deep learning models for diabetes risk prediction. In *Internet of Things and Machine Learning for Type I and Type II Diabetes* (pp. 135-146). Elsevier.

IEEE CONFERENCE:

- Yadav, K., **Tanwar, N., & Hasija, Y.** (2024). Exploring the role of gut microbiota in hypertension: Insights from Machine Learning and eXplainable AI. *15th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, June 2024.
- Tanwar, N., Meena, J., & Hasija, Y.** (2022). Explicate Toxicity by eXplainable Artificial Intelligence. *International Conference on Industry 4.0 Technology (I4Tech)*, September 2022.
- Tanwar, N., & Hasija, Y.** (2022). Explainable AI: Are We There Yet? *IEEE Delhi Section Conference (DELCON)*, February 2022.
- Tanwar, N., & Hasija, Y.** (2022). Deep Learning: A Tool in Biomedical Science. *International Conference for Advancement in Technology (ICONAT)*, January 2022.

INTELLECTUAL PROPERTY

Design Patent (Under Consideration)

Portable Food Spoilage Detection Device – Dual-sensor (gas + TDS/pH) system for real-time detection of food spoilage.

Application No.: **461910-001**

Status: FER Review (2025)

Nakul Tanwar

EDUCATION

- **Ph.D. Scholar- Department of Biotechnology, Delhi Technological university**
2022- 2026

Thesis Title: *An Integrated Approach Towards the Identification of Novel biomarkers in Respiratory Disorders* (Supervisor: Prof. Yasha Hasija)

- **MSc Biotechnology- Delhi Technological University**
2020-2022

Project Title: *Explaining Developmental Neurotoxicity By XAI* (Supervisor: Prof. Yasha Hasija)

- **B.Sc. (Honors) Biotechnology- Amity University, Gurugram** 2017-2020

WORK EXPERIENCE

- **Delhi Technological University / 2022–2025 / Teaching Assistant**

Guided B.Tech and M.Tech students in bioinformatics, machine learning, and multi-omics projects, assisting in data analysis, model interpretation, and research documentation.

- **Imperial life sciences/ MAY-JULY 2021/ Lab Intern**

Assisted in preparing and validating RT-PCR kits for COVID-19. Gained hands-on experience in molecular diagnostics, including process optimization, reagent prep, and quality control- contributing to public health efforts.

- **Janakpuri Super Specialty Hospital/ May- July 2019/ Lab Intern**

Worked in hematology and pathology labs, performing CBC analysis, blood smears, and histopathology preparation. Built strong foundations in diagnostic techniques, reagent handling, and equipment maintenance.

SKILLS

- **Python:** Strong proficiency in scientific computing, data processing, automation, and model development using NumPy, Pandas, scikit-learn, and visualization libraries.
- **Machine Learning:** Hands-on experience with supervised and unsupervised models, feature engineering, model evaluation, and explainable AI techniques (SHAP/LIME).
- **Data Analysis:** Skilled in exploration data analysis, statistical analysis, data cleaning, and translating insights into actionable outcomes.
- **LLMs & NLP:** Design and deployment of LLM-augmented pipelines for biomedical text understanding, risk stratification, and explainable ML workflows; experience with prompt optimization, retrieval-augmented generation (RAG), and model evaluation.

ACHIEVEMENTS & AWARDS

- Commendable Research Award, DTU (Research & Innovation Excellence Awards 2025)
- GIAN Course: *Medical Translational Research – Bench to Bedside* (2025), Prof. Hitesh Handa